

The Link: Wages, Machines, and What Remains

A task model of automation, job rents, and the fiscal system, with U.S. measurements, 1962–2025

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Abstract. Wages are prices on the *link*: the non-uniformity of human capability relative to machines across tasks. Pay decomposes into three strata: an exit floor; a premium set by the level of the capability schedule at the marginal task and insulated by its slope; job-specific rents on top. Automation demolishes the strata from the top down: profit-directed adoption targets the best-protected jobs first (Acemoglu and Restrepo, 2026), then flattens the schedule, then crosses the participation margin. When machines make machines, the value released migrates to rents on non-produced factors: land, sites, energy. The real wage in machine-made goods is bounded below by absolute human productivity; the collapse runs through the land-priced components of consumption — immiseration through rent, not through the wage. Roughly seven-tenths of U.S. consumption and revenue are wage-linked, so the fiscal system sits downstream of the collapsing base. Conditional transfers fail in regime-specific ways; an unconditional transfer funded by rents on non-produced factors is the unique fiscal instrument whose base cannot contract and whose payment moves no margin — unique up to an equivalence between consumption and rent bases that becomes exact in the full-automation limit the paper calls the corner. Its coverage ratio — site rents against a universal subsistence floor — is measured at one-third and rising. At the corner the tax-and-dividend pair delivers what the second welfare theorem promises — redistribution without deadweight — with instruments that exist: the corner's character is institutional, not technological. The paper states the model with proofs, reports U.S. measurements for 1962–2025, and lists thirteen falsifiable predictions.

1. The claim

Begin with the classical economists. Ricardo (1817) priced land at the margin: each parcel earns its advantage over the worst in cultivation. The wage he tied to subsistence — the natural price of labor — Malthus (1798) supplying the mechanism: population absorbs any surplus above it. Profits take what remains. George (1879), writing six decades later, outside the tradition, pushed the machinery one turn further: growth settles in rent, not in wages. None of them thought in tasks; a tool was just capital. Ricardo himself broke ranks late: the machinery chapter added in 1821 concedes that machinery can injure the laboring class. The concession stayed a chapter; here it becomes the model. Recast in this paper's terms, their shared premise: the human edge over tools varied little from task to task. Wages at subsistence, rent collecting the growth — not a naive picture, but bookkeeping for the world they had.

The industrial revolution broke the premise. Machine capability split from human capability, violently and unevenly: engines far better than muscle at the physical tasks, useless at the cognitive ones. Labor was valuable exactly where machines were not; the wage became a price on

the gap. Two centuries of rising wages made the classical laws look refuted: land dropped out of the production function, and the subsistence wage passed into the history of thought. This paper reads the record differently: the laws were not wrong; they were conditional. The premise stopped holding; the conclusions went dormant with it.

AI is the candidate reversal: machine capability improving across tasks, not one family of them. Whether the non-uniformity closes is an empirical premise, not a definition; Section 10 states it plainly; Section 9's predictions put it at risk. Machines and humans can each do most tasks; what differs is by how much. Where the difference is large and uneven, labor complements machines and commands a premium far above subsistence. Where capability becomes uniform, the premium evaporates — not because humans got worse at anything, but because the gap closed. If the closing continues, the premise returns, and the dormant conclusions wake with it.

One disclosure belongs up front. The core idea and its central claims are the author's. The formalization, the proofs, the computations, and most of these sentences were produced by an AI system under her direction and review. No proposition entered the draft until its algebra passed a computer-algebra check and its equilibrium claims a numerical one. It is stated here because the claim that capability is becoming uniform is not a speculation about other people's occupations: the capability priced in this paper is the capability that drafted it.

What follows is the bookkeeping for that return: the classical laws inside a modern task model with a fiscal state attached. This paper takes one sentence seriously and follows it to the end: **your wage is the rental price of a machine, times your edge over it, at the task where your edge is smallest among the tasks humans still hold.** Not need, not effort, not merit: replacement cost at the margin.

Three consequences follow, each developed below with proofs and then measured. First, a demolition schedule: automation removes the layers of a paycheck in a fixed order — job rents first (and the better protected the job, the earlier the robot comes for it), the capability premium second, participation last — with distinct observable signatures at each stage, one of which has already been found in U.S. data.

Second, a conservation law: when machine-making itself automates, machine prices collapse to their land-and-energy content, so the value released by the demolition reattaches to non-produced factors; "rent dissipation" is really rent migration. A corollary answers the classic objection to immiseration stories: machine-made goods do get cheaper, so the squeeze arrives through the land-priced part of the budget, not the goods part.

Third, a fiscal theorem: the tax-and-transfer state is plumbed almost entirely into the wage loop — funded by wages, keyed to work — so it thins as the loop does; in the limit, redistribution funded by taxes on wages becomes an exact shell game, while a uniform transfer funded by a tax on site rents is the one instrument that survives — not as a preference among options, but as the only column of the ledger with anything left in it.

Two further results close the arc. A feasibility theorem prices the remedy: the U.S. coverage ratio of a fully rent-funded floor measures one-third today and climbs with automation itself. The strongest objection, Baumol concentration on human-essential tasks, is granted its mechanism and priced: it forks on substitutability, binds only where humanity can be verified (in law or in person), and where it rescues the aggregate labor share it leaves the median wage to collapse anyway. A closing welfare theorem draws the moral: at the corner — the full-automation limit — the land-tax-and-dividend pair delivers redistribution without deadweight, the second welfare

theorem's promise, with instruments that exist. The corner's character, dystopia or inheritance, is a choice of tax base, not of technology.

Related work

The machinery is Acemoglu and Restrepo's task framework (2018, 2022), run with the new-task margin closed — Section 10 defends the closure — and their rent-dissipation mechanism (2026) transplanted onto the capability schedule. Korinek and Stiglitz (2019) posed the below-subsistence scenario this model resolves through exit; Korinek and Suh (2024) is the closest formal predecessor on wage collapse under transformative AI, differing here in where the collapse ends — at non-produced factors rather than capital accumulation — and in the conditionality contrast of Propositions 6 and 7. Susskind (2017) and Sachs and Kotlikoff (2012) model technological unemployment and immiseration through other channels; Moll, Rachel, and Restrepo (2022) route automation's gains to wealth-holders, where this model names the terminal claimant: land. Caselli and Manning (2019) prove real wages cannot fall in terms of machine-made goods — the objection Proposition 4 concedes and then answers through the land content of subsistence. Grossman and Rossi-Hansberg (2008) supply the trading-tasks lens; here the foreign wage enters as a second machine rental, so offshoring and automation are one demolition read at two heights.

Zeira (1998) is the replacement ancestor; Autor (2015) the reinstatement-side counterpoint; Hémous and Olsen (2022) endogenize automation's direction — the engine the fortification remark of Section 4 leans on; Karabarbounis and Neiman (2014) document the decline's global reach. Aghion, Jones, and Jones (2019) supply the strongest stabilizer — Baumol concentration of expenditure on human-essential tasks — engaged in Section 10. George (1879) supplies the terminal instrument; Ricardo (1817), the margin logic the whole model runs on.

2. The model

Tasks. Producing the final good requires a continuum of tasks $x \in [0,1]$, each in equal amount — a checklist where every box is ticked once per unit of output (Leontief aggregation over tasks). The good's unit cost is the sum of its tasks' unit costs.

Two ways to tick a box. One human-hour completes $\gamma_L(x)$ units of task x ; one machine-hour completes $\gamma_M(x)$ units. The *human edge* is $\rho(x) = \gamma_L(x)/\gamma_M(x)$. A machine-hour rents for c — taken as given until Section 5, where it becomes the second protagonist.

Wedges. Some jobs pay more than the market requires: union premia, licensing rents, efficiency wages paid because monitoring is hard. Following Acemoglu and Restrepo (2026), model these as a wedge $\mu(x) \geq 1$ that firms must pay on task x , with $\mu = 1$ on a positive mass of tasks. Wedge jobs are rationed: firms hire into them only up to the point where the inflated wage equals marginal product, and workers queue for them — a convention that returns in Section 3: because the rationed jobs are full, the marginal participant is evaluated at the base wage.

People. N identical workers each supply one hour or exit to an outside option worth s — the woods, the family, the commons: the value of the best life available *without* selling labor. Note that s quietly presupposes access to some land — a debt Section 7 prices.

Land. Fixed in total, heterogeneous in quality, worst parcel's reservation rent zero. Idle until Section 5, decisive after.

Lemma 1 (Deflation). Labor holds task x iff $\mu(x) \cdot w / \gamma_L(x) \leq c / \gamma_M(x)$, i.e. iff $w \leq c \cdot \tilde{\rho}(x)$, where $\tilde{\rho}(x) = \rho(x) / \mu(x)$. All allocation in the model runs on the wedge-deflated schedule $\tilde{\rho}$, not the raw one.

Proof. Divide both sides of the cost comparison by $\mu(x)$. ■

Geometrically: **a wedge is negative altitude.** It drags a task down the effective schedule toward the waterline — the height w/c below which machines hold every task — without changing how good humans actually are at it. For the threshold argument, relabel tasks so $\tilde{\rho}$ increases from left to right, and assume the relabeled schedule is continuous and strictly increasing where labor is employed; Figure 1 keeps the raw ordering so the pockets stay visible. Competitive firms assign each task to the cheaper input, so there is a threshold x^* : machines take $[0, x^*)$, labor takes $[x^*, 1]$, and in equilibrium firms are exactly indifferent at the margin while everyone willing to work at the going wage is working or has exited.

Lemma 2 (Existence and uniqueness). With the relabeled $\tilde{\rho}$ continuous and strictly increasing where labor is employed, an equilibrium pair (w, x^*) exists and is unique for any number of participants; on a flat stretch the wage is pinned and only the assignment of identical-cost tasks is indeterminate, with nothing riding on it.

Proof. Absorbing more participants requires labor to hold more tasks, so the market-clearing wage $c \cdot \tilde{\rho}(x^*)$ (n)) is continuous and strictly decreasing in the participant count; participation supply is a step at s . A strictly decreasing continuous curve crosses a step exactly once, with the exit resolution of Proposition 1(iii) handling the below- s branch; flat stretches pin the wage at $c \cdot \tilde{\rho}$ and leave free only a measure-preserving relabeling of identical tasks. ■

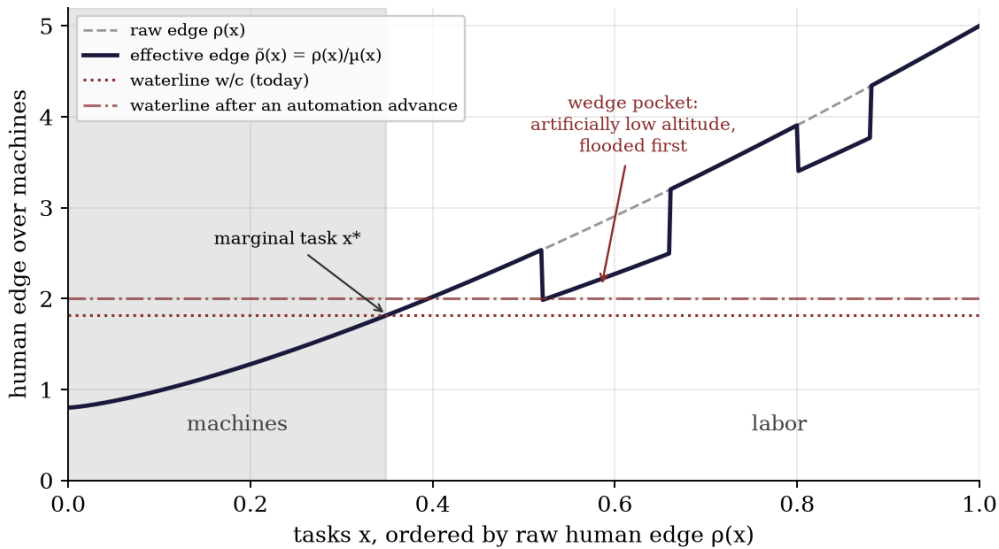


Figure 1. The task space. Tasks are shown in raw-edge order; the dashed gray curve is $\rho(x)$, and the solid effective edge $\tilde{\rho}(x) = \rho(x) / \mu(x)$ sits below it wherever wedges bite. The threshold argument relabels by $\tilde{\rho}$. Machines hold everything below the waterline w/c ; labor holds everything above. Wedge pockets sit below their raw altitude — and when the waterline rises with an automation advance, the pocket floods first even though the humans in it are objectively better at their task than neighbors who survive.

3. The wage is a price on the link

Proposition 1 (The price on the link). (i) In any equilibrium using both inputs, the base wage is $w = c \cdot \tilde{\rho}(x^*)$. (ii) The wage's sensitivity to human scarcity is governed by the slope of $\tilde{\rho}$ at x^* : steep slope, responsive and insulated wage; flat schedule $\tilde{\rho} \equiv \bar{\rho}$, labor demand perfectly elastic at $c \cdot \bar{\rho}$ and the wage pinned there regardless of how many or few humans exist. (iii) No equilibrium wage sits below s while anyone works; if the clearing wage would fall below s , workers exit until the wage rises to s (possible only when the schedule has slope) or all have left.

Proof. (i) If $w > c \cdot \tilde{\rho}(x^*)$, the marginal task and, by continuity, tasks just above it are cheaper by machine; firms drop labor there; the unemployed underbid; w falls. If $w < c \cdot \tilde{\rho}(x^*)$, tasks just below x^* strictly favor labor; firms demand more hours than exist; bidding raises w . Only equality is stable. (ii) To absorb more hours, labor must win more tasks, and a firm hands over a task only when the wage falls relative to that task's machine cost; so more participants push x^* left and w down, in proportion to the slope. Flat schedule: no wage above $c \cdot \bar{\rho}$ employs anyone, every wage at $c \cdot \bar{\rho}$ employs everyone willing. (iii) is the participation condition applied to (i) and (ii). ■

Definitions. *The link* is the pair (level, slope) = $(\tilde{\rho}(x^*), \tilde{\rho}'(x^*))$. The level sets the wage; the slope insulates it from scarcity. Call the schedule-has-slope case the *linked regime* and the flat case the *corner regime* — labor demand perfectly elastic at machine parity, the pinned wage $c \cdot \bar{\rho}$ at which labor and machine cost the same at every task — with two sub-cases: corner-above ($c \cdot \bar{\rho} \geq s$ — everyone works at a pinned wage) and corner-below ($c \cdot \bar{\rho} < s$ — rational exit). The corner-below case resolves the "wages below subsistence" scenario of Korinek and Stiglitz (2019) the only way it can be resolved: by leaving, not by dying — in practice, by becoming a dependent, because a mother can see value in her child that the economy cannot.

Running example. $c = \$10$, $s = \$25$. Task A: $\rho = 1$, $\mu = 1$ (machines hold it). Task C: $\rho = 4.5$, $\mu = 1$. Task B: $\rho = 5$, $\mu = 1.25$ — objectively humans' best task, but wedged. Effective edges: $\tilde{\rho}(B) = 4$, $\tilde{\rho}(C) = 4.5$, so B is the effective marginal task and $w = \$40$. The C-worker earns \$40; the B-worker earns $\mu \cdot w = \$50$. Now let AI improve machines uniformly by 25%: raw edges fall to $\rho(B) = 4$ and $\rho(C) = 3.6$. If nothing else changed, the new wage at the B-margin would be \$32 — a 20% pay cut delivered entirely by someone else's machine getting better at *your* task, with no human getting worse at anything. One caveat carries its own proposition: this arithmetic is nominal — it holds the machine rental c fixed while machines improve, and machine progress also cheapens machine-made goods. Proposition 4 does the general-equilibrium bookkeeping and locates exactly which part of the cut is real.

History in one figure. Before the industrial era, machines scarcely existed; wages sat near s , pinned by land scarcity — Ricardo's version of the same margin logic, run through soil instead of silicon. The industrial revolution split capabilities violently: machines absurdly better at the physical, useless at the cognitive. The schedule became enormously dispersed and steep, the link tight, wages high and insulated — the historical anomaly we mistake for a law. Computing began flattening the simple-cognitive end. AI makes the flattening general, and Figure 2's annotation is the whole argument: the level falls and the slope falls, and they are separate injuries. Technology giveth, and technology taketh away.

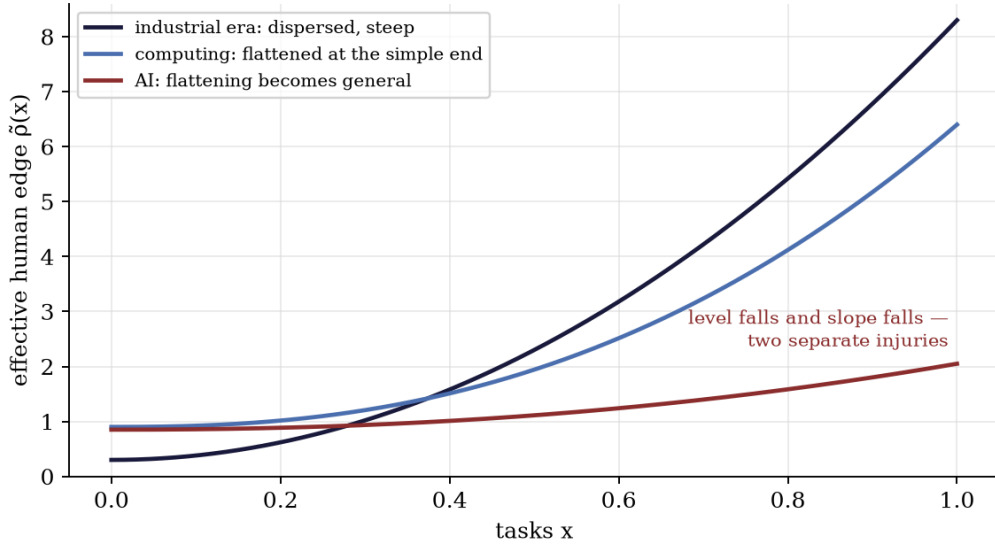


Figure 2. The capability schedule across three technological eras (the pre-industrial case is off-chart: almost no machine tasks at all, with the wage floor enforced through the land margin instead). Flattening lowers the wage through the level and strips its insulation through the slope.

Heterogeneity. Real people carry personal schedules and personal outside options. Every result below holds worker-type by worker-type: each type prices its own link, so heterogeneity lifts the mean above the floor — the cross-sectional shadow of Proposition 1. Floor statements refer to the bottom of the support, not the average.

4. Demolition from the top

A paycheck now has three geological layers, and the model's central narrative is that automation removes them in order.

$$\text{wage at task } x = s + [c \cdot \tilde{p}(x^*) - s] + [\mu(x) - 1] \cdot c \cdot \tilde{p}(x^*)$$

(exit floor) + (link premium) + (wedge rent)

Proposition 2 (Targeting; Acemoglu–Restrepo 2026, transplanted). When an automation advance makes a set of tasks feasible for machines, the tasks actually flipped are those with the lowest $\tilde{p}$ among the feasible ones. The adoption region is a lower set in the effective schedule, so a wedge raises its task's flip priority one-for-one; under Acemoglu and Restrepo's (2026) conditions on the advance — no bias toward low-wedge tasks — the flipped set is, in their distributional sense, the high- μ set: **automation is targeted at the best-paid versions of each job.** The proviso is load-bearing: a wedge whose institution can reach the feasible set itself — blocking adoption by law or contract rather than pricing labor — exits the theorem's domain; the fortification remark below tracks where it goes.

Proof. Adoption occurs where labor cost exceeds machine cost, $\mu(x) \cdot w / \gamma_L \geq c / \gamma_M$, i.e. where $\tilde{p}(x) \leq w/c$: the adoption region is a lower set in the effective schedule, and wedges are what push otherwise-high tasks into it. ■

In the running example: after the 25% machine improvement, $\tilde{p}(B) = 3.2$ and $\tilde{p}(C) = 3.6$, so **B floods before C — automation takes the task where humans are objectively better** ($4 > 3.6$), because the wedge made it the cheapest conquest. The displaced worker loses twice, and

differently: the \$10 wedge rent vanishes with full incidence — a composition effect that moves no margin, so nobody shares it — while the base-wage decline is spread economy-wide through the margin. Acemoglu and Restrepo find exactly this asymmetry empirically: displacement effects propagate across groups; rent losses land undampened on the displaced.

Three corollaries, each a prediction:

Self-liquidation of labor-side protection. A wedge that operates through price raises its own task's automation priority. Union premia, licensing rents, and any legally mandated "human-made" price premium are self-undermining under sustained automation; the only protection that survives Proposition 2 is quantity exclusion — law removing a task from the automatable set entirely. The better protected the job, the earlier the robot comes for it.

The within-group signature. Because wedge jobs sit in the upper-middle of each group's internal wage distribution, stage one of the demolition *compresses* wages within exposed groups: losses concentrate in roughly the 70th–95th within-group percentiles, with the bottom (no rents to lose) and the very top (unautomatable rent jobs) moving only with the base wage (Figure 4). Acemoglu and Restrepo (2026) document precisely this U-shape for U.S. groups, 1980–2016 — a finding that contradicts the folk assumption that technology always widens inequality at every level of aggregation. This model inherits its mechanism from that finding — calibration, not confirmation; the predictive content is the pattern's persistence, and its inversion where protection holds the adoption decision (Predictions 1–2).

The subsidy mirror. An in-work benefit is algebraically a wedge $\mu < 1$ — labor accepting less at a task — producing over-employment where private rents produce under-employment. Private and public distortions are the two half-lines of a single μ -axis, with the competitive model at $\mu = 1$; Section 6 returns to the public half.

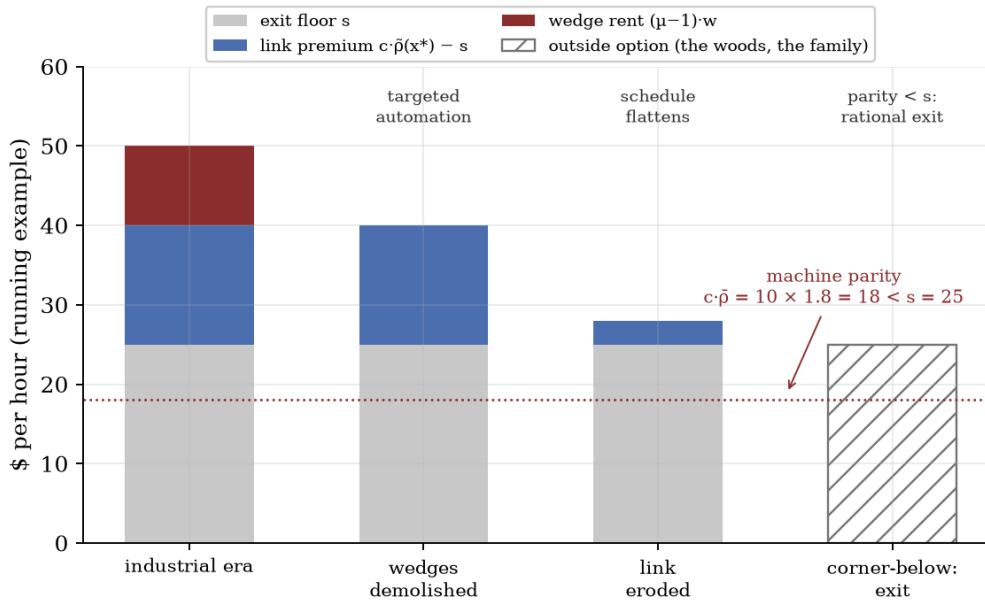


Figure 3. The demolition schedule, in the running example's dollars. Wedge rents die first (Proposition 2 targets them); the link premium erodes second as the schedule flattens; participation goes last, when machine parity crosses the outside option and the rational move is exit — in practice, dependency. Each stage has a distinct empirical signature. Terminal-stage numbers: $c = \$10$, $\bar{p} = 1.8$, $s = \$25$.

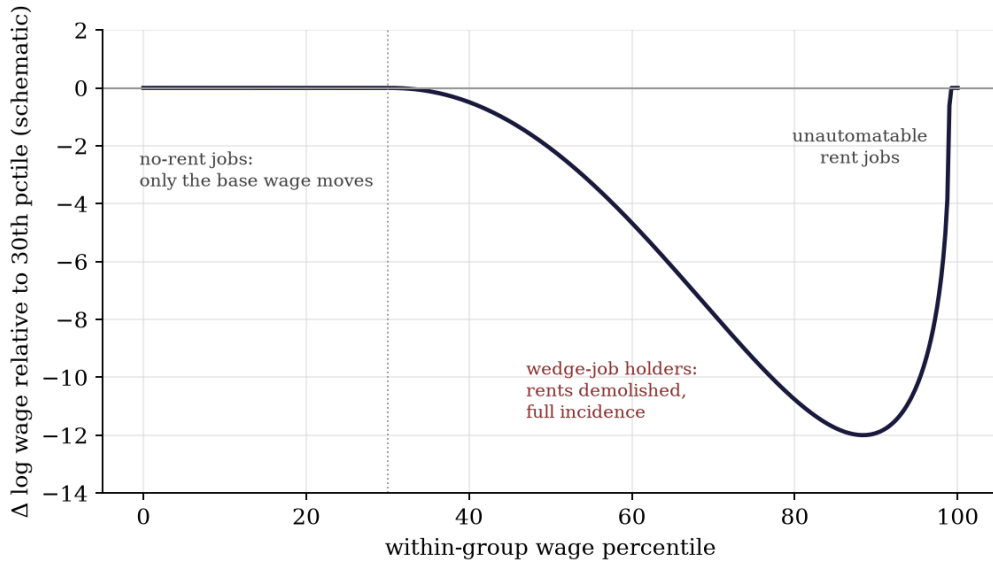


Figure 4. Predicted change in wages across the *within-group* distribution for a group exposed to automation (schematic). The U-shape is the fingerprint of stage-one demolition: rent-holders lose their stratum with full incidence while everyone else moves only with the base wage. Acemoglu and Restrepo (2026) estimate this exact pattern in U.S. data, 1980–2016, with automated jobs estimated to pay rents on the order of 40–50% above base wages.

The transition can be negative-sum. Wedges break the comforting theorem that automation, however brutal distributionally, always raises total output. A firm automating a \$50 wedge job saves \$50 minus the machine; society only frees what the worker produces elsewhere — \$40 minus the machine. The \$10 gap is a pure allocative loss, incurred every time Proposition 2's targeting fires. Acemoglu and Restrepo estimate that for the U.S., 1980–2016, this loss offset 60–90% of automation's productivity gains — possibly all of them. Privately profitable automation ran close to socially break-even, and perhaps negative-sum, *before any of this model's later concerns arrive*. The loss has a finite fuel tank — it is capped by the wedge stock, and once the wedge stratum is demolished the competitive envelope logic resumes — but "finite" here has already meant four decades.

Remark (fortification: what the demolition leaves standing). The natural objection to Proposition 2 is that the best-protected workers will not wait to be repriced: they will block the machine. The objection is correct, and the theorem has already marked its boundary. Proposition 2 governs wedges that operate through the price of labor: the firm pays more but still chooses freely between labor and machine. Some wedge-holders can reach the feasible set itself: a law or contract can remove the machine option outright rather than make labor dear. That power sits outside the theorem's domain — the proviso's case. Call such a task *fortified*.

Not every protection can fortify. Bargaining power cannot, not durably. A strike threat is priced leverage: its value is the cost of replacing the strikers. Automation feasibility is precisely a fall in that cost, so bargained protection declines on the same schedule as the wage it defends. History confirms it: dock mechanization agreements and print-room lifetime guarantees were vetoes sold — blocking power converted into ordinary transition rents. What fortifies is a rule: a scope-of-practice requirement extended to software, an automation ban in an enforceable contract, a mandated human in the loop. These survive machine parity, because no improvement in γ_M repeals a rule. The sold vetoes were themselves contracts, the weaker rule form: a contract

fortifies only until expiry, when renewal must be won with decaying bargaining power. A statute never expires; what it protects, only politics can dislodge, at a price.

Under demolition pressure, protection converts from price form to quantity form, and the demolition acts as a filter: it removes price wedges in $\tilde{p}$ -order and leaves an increasingly law-anchored residue. Three consequences. (a) Adoption at fortified tasks is punctuated. Nothing moves while the firm's per-period saving from flipping runs below the cost of confrontation — the legislative fight, the contract's expiry. When it clears the cost, every task the rule covers flips at once. A merely priced wedge flips smoothly, at gap zero.

(b) The "finite fuel tank" caps only the priced stratum Proposition 2 can reach. A blocked task's allocative loss (the worker valued at the base wage, as in the negative-sum accounting) is $w/\gamma_L - c/\gamma_M$ per task unit. As machine costs fall it grows toward the full labor cost. Fortified tasks concentrate in sheltered, non-tradable sectors. Where machine-made goods substitute poorly for those sectors' output, their expenditure share grows as goods cheapen — the Baumol side of the budget, priced in Section 10. The drag compounds on both margins.

(c) A fortified wage is institutional rent in wage form: inframarginal, taxable like rent, revocable like law. Section 5's accounting keeps a column open for it. (The manager — wedge and adoption decision in one chair — is the limiting case, taken up in the appendix.)

The demolition dissolves every protection written in prices; what is left standing afterward is written in law.

5. Where value goes when machines make machines

So far the machine rental c has been a given. This section makes it the second protagonist: when machines are made by machines, competition among machine-makers sets what c can be, and the answer decides where the value released by Section 4's demolition comes to rest. The proposition prices a machine-hour from its recipe.

Proposition 3 (Value collapse). Let producing one unit of machine services require a units of machine services ($a < 1$: machines net-reproduce) and ℓ units of non-produced services — sites, energy, ore — renting at r . Free entry forces $c = a \cdot c + \ell \cdot r$, hence

$$c = \ell r / (1 - a).$$

With time preference δ (building takes a period), $c = \ell r(1+\delta)/(1 - a(1+\delta))$, finite iff $a(1+\delta) < 1$; the wedge δ is interest — the wage of waiting — competitive, real, and thin. Durability generalizes it: with wear at rate d the user-cost recursion gives $c = \ell r(\delta+d)/(1 - a(\delta+d))$ — the display is the one-period case $d = 1$ — so wear thickens the interest sliver as waiting does, and the destination is unchanged.

Proof. Above recipe cost, entrants build machines with machines and undercut; below it, builders exit. The displayed equation is the zero-profit condition; solving it is one step. (With many machine types this is a matrix equation $c = Ac + Lr$ solved through $(I-A)^{-1}$; the scalar carries the full logic.) ■

Corollary (the surviving-income taxonomy). In the corner regime, gross income decomposes exhaustively into rents on three kinds of scarcity — *physical* (land, energy, sites, water: permanent), *institutional* (whatever law elects to protect: IP, licenses — and by Section 4's self-liquidation result, only quantity exclusions endure), and *dynamic* (transition rents while

capacity builds: real, temporary, and "temporary" can mean decades for a nuclear plant) — plus thin interest. Wages have left the list.

Ideas, the boundary case. An idea's reproduction requires approximately nothing of anything, so its competitive price is approximately zero; any positive price for an idea is institutional scarcity, chosen rather than natural. Chosen, as Romer (1990) made canonical, for a reason: at marginal-cost pricing, ideas with fixed creation costs are not created at all, so the institutional scarcity is constitutive of the idea sector, not a gratuitous distortion. Which ideas earn rents, and for how long, remains a policy dial; and once an income floor exists, it is itself a second margin for funding creation.

Worked instance. Model weights are just numbers — an idea. Distillation pierces even trade secrecy, because the behavior leaks out through the product's own outputs; no patent regime fully contains a function you can query. The ownable things are, surprise, the datacenters, the land under them, the power contracts, and the water rights. Proposition 3 says the market value of the AI sector migrates from the former to the latter — whatever the current market believes.

Proposition 4 (Real wages and the land content of subsistence). In the corner regime, with the recursion of Proposition 3: (i) the real wage in machine-produced goods is $w/p_g = 1/\bar{L}$, where $\bar{L} = \int dx/\gamma_L(x)$ is the labor a human would need to complete the whole checklist alone — absolute solo productivity, invariant to machine quality; the machine rental cancels. (ii) The price of non-produced services relative to machine-made goods is $r/p_g = (1-a)/(\ell \cdot \bar{p} \cdot \bar{L})$, and it diverges as the task-side edge closes ($\bar{p} \rightarrow 0$) or the recipe's land content vanishes ($\ell \rightarrow 0$). The self-replication margin moves it boundedly, and in the opposite direction to a naive reading: improvement is a fall in a , which raises r/p_g toward its bound $1/(\ell \cdot \bar{p} \cdot \bar{L})$, while $a \rightarrow 1$ is regress — it sends machine costs, not rents, to infinity ($c = \ell r/(1-a)$), and $r/p_g \rightarrow 0$. Even substituting machines for land inside the recipe, $\ell = \ell_0(1-a)$, leaves $r/p_g = 1/(\ell_0 \cdot \bar{p} \cdot \bar{L})$: bounded. (iii) Against a consumption bundle with expenditure share σ on non-produced services — housing, siting, energy, food-acreage — the real wage collapses if and only if $\sigma > 0$; at $\sigma = 0$ it is bounded at $1/\bar{L}$. The participation constraint $w \geq s$ binds through the land-priced component of subsistence.

Proof. (i) With ρ flat at $\bar{\rho}$, $\gamma_M(x) = \gamma_L(x)/\bar{\rho}$, so $p_g = c \cdot \int dx/\gamma_M = c \cdot \bar{\rho} \cdot \bar{L} = w \cdot \bar{L}$. (ii) Substitute $c = \ell r/(1-a)$ into $r/p_g = r/(c \cdot \bar{\rho} \cdot \bar{L})$. (iii) A geometric price index $P = p_g^{1-\sigma} \cdot r^\sigma$ gives $w/P = (1/\bar{L}) \cdot (p_g/r)^\sigma$, which by (ii) tends to zero along the divergent margins ($\bar{p} \rightarrow 0, \ell \rightarrow 0$), for any $\sigma > 0$. ■

This concedes and then answers the strongest objection to every immiseration story (Caselli and Manning, 2019): cheaper machines cannot starve anyone *in terms of the things machines make* — correct, and beside the point, because subsistence is not made of machine-made goods alone. Space, shelter, energy, and the acreage under food carry irreducible non-produced content, and that is precisely where Proposition 3 sends the value. **Immiseration, when it comes, arrives through rent, not through the wage:** the pay packet buys as many gadgets as ever and ever less ground. The exit option s was land-backed from Section 2; the theory closes on itself.

Closing the loop: where the spending goes

Proposition 3 prices machines from the cost side; nothing so far says where household spending comes to rest. This subsection closes the corner-regime economy and states "conservation of

rents" as what it is at the corner: an accounting identity.

In words, before the algebra: at the corner, every dollar of household spending ends as rent on some site — goods spending traced through the machine sector's zero-profit condition, housing spending directly. Land is the only primary factor still employed, so the flow of spending has nowhere else to come to rest. It follows that national income and aggregate site rent are the same number. Proposition 5 also recovers from the demand side the relative price r/p_g that Proposition 4(ii) derived from the cost side. With this technology prices are cost-determined, so the demand-side derivation is not a test the model could have failed; what it shows is that the bookkeeping closes — no value leaks, and none appears from nowhere.

Setup. Households jointly hold the land (the distribution of claims is arbitrary here, and becomes the subject of Proposition 8). Preferences are Cobb–Douglas over the machine-made final good g and direct land services h — housing, sitting, space — with land share $\sigma \in (0,1)$: $U = g^{1-\sigma} \cdot h^\sigma$. Land is homogeneous with total stock T , renting at r , split between production use T_P (entering machine-making through ℓ) and housing T_H . Machines perform all tasks (labor has exited, or is of negligible measure); $\delta = 0$, so the interest sliver is suppressed — with $\delta > 0$ a thin interest flow reappears on the income side and nothing below changes beyond carrying the term. Free entry in machine-making gives $c = \ell r / (1-a)$ and hence $p_g = c \cdot \bar{p} \cdot \bar{L}$.

Proposition 5 (Closure and conservation of rents). In the corner regime: (i) equilibrium exists and the real allocation is unique,

$$T_H = \sigma \cdot T, \quad T_P = (1-\sigma) \cdot T, \quad Y = (1-a)(1-\sigma) \cdot T / (\ell \cdot \bar{p} \cdot \bar{L});$$

(ii) the entire goods bill is production-land rent, and national income is aggregate site rent identically,

$$p_g \cdot Y = r \cdot T_P, \quad p_g \cdot Y + r \cdot T_H = rT;$$

(iii) market clearing reproduces the cost-side relative price of Proposition 4(ii), $r/p_g = (1-a)/(\ell \cdot \bar{p} \cdot \bar{L})$: demand side and cost side close on the same price.

Proof. Gross machine services X solve $X = \bar{p} \cdot \bar{L} \cdot Y + a \cdot X$: services for final tasks plus services consumed making services. Hence $X = \bar{p} \cdot \bar{L} \cdot Y / (1-a)$, and production land demand is $\ell \cdot X$. Substituting $p_g = \ell r \cdot \bar{p} \cdot \bar{L} / (1-a)$ gives $p_g \cdot Y = r \cdot \ell \cdot \bar{p} \cdot \bar{L} \cdot Y / (1-a) = r \cdot \ell \cdot X = r \cdot T_P$. Goods spending passes through the zero-profit machine sector and lands, all of it, on production land. Land is the only primary factor employed, so household income is rT . Cobb–Douglas shares give $r \cdot T_H = \sigma \cdot rT$ and $p_g \cdot Y = (1-\sigma) \cdot rT$, hence $T_H = \sigma \cdot T$; land-market clearing $T_P = T - T_H$ then pins Y as displayed; and dividing $p_g \cdot Y = (1-\sigma) \cdot rT$ by the expression for Y returns $r/p_g = (1-a)/(\ell \cdot \bar{p} \cdot \bar{L})$, which is (iii). Uniqueness is immediate from the shares. ■

Dissipation is migration, by construction: every unit of final spending resolves through the machine recursion into a claim on a site. The obvious escape route is that automation's released value might dissipate into consumer surplus at collapsing prices rather than reattach to land. That route is closed. Prices do collapse *and* the identity holds, because what collapses is p_g relative to r , never the rent's claim on the flow of spending. The wealth data already lean this way — Rognlie (2015) shows the entire postwar rise in the net capital share is essentially housing, which is to say land.

Three remarks are deferred to the appendix — the CES dial between goods and land services, whose elasticity η Prediction 8 reads off the data; heterogeneous parcels with an idle margin; and the corner-above case with labor still present. None changes where anything goes.

Remark (distribution, made stark). The identity prices the stakes of Proposition 7: in the corner, a household without land claims has zero market income — not low; zero. Whether rents can fund the floor is therefore not one policy question among several; it is the question of whether anyone without a site eats. Proposition 8 takes it up.

6. The fiscal system sits downstream of the wage loop

This section asks how much of the state's own plumbing runs through the wage — and what becomes of it as the wage goes. For each fiscal instrument i , let λ_i be the wage-linked share of its base — for transfers, of its funding or its eligibility, as the question requires. Four accounting facts follow from Propositions 1 and 3. (i) In the corner regime the current-wage base shrinks toward participation $\times$ parity — toward zero in corner-below — so λ -weighted revenue contracts in proportion. (ii) The eligibility scissors: work-conditioned claims contract with participation while need-based claims expand; revenue and obligations cross like blades. (iii) Instruments taxing *embodied past labor* — the structures share of property tax, consumption financed from past-wage savings — keep yielding through the collapse and decay only at depreciation and drawdown rates: the fiscal cliff has a time constant, measured in years to decades. Cushion, not comfort. (iv) VAT is the exception that matters: its base survives, and its incidence migrates by itself to whatever finances consumption — which in the corner regime, by Proposition 3, is rents. The one instrument that follows the value without being redesigned.

Proposition 6 (Conditionality). The proposition runs the three ways a transfer can be conditioned — on nothing, on working, on not working — through the participation margin. (i) An unconditional transfer u enters both sides of the participation condition — work iff $w + u \geq s + u$ — and cancels: no margin moves, while the disposable floor rises by exactly u . The cancellation decomposes: u leaves the work–exit price $w - s$ untouched, so its compensated (substitution) response is zero as an identity; what remains is income effect. Once $u \geq s$, exit is feasible without woods or family: the transfer is a purchased commons, re-providing the outside option that enclosure priced away.

(ii) An in-work benefit b lowers the reservation wage to $s - b$. In the linked regime, the induced participation pushes the margin left and part of b leaks into lower wages — the Speenhamland result rebuilt without employer power, the leak flowing through the market's own margin and growing as the schedule flattens. In the corner regime the wage is pinned, so there is no leak — but if $s > c \cdot \bar{p} \geq s - b$, the benefit summons work whose replacement value is $c \cdot \bar{p}$ from people whose time is worth s outside: society burns $s - c \cdot \bar{p}$ per hour to preserve the appearance of employment. The failure mode is regime-dependent: leak when linked, make-work in the corner.

(iii) An out-of-work benefit b' raises the reservation to $s + b'$ and taxes the first hour of work at b' ; as parity compresses the wage distribution toward the floor (Section 4, stage one), the mass of workers inside any fixed withdrawal band rises mechanically. The same rulebook generates a worsening trap; the dysfunction belongs to the regime, not the rules.

Proof. The displayed inequalities plus Proposition 1's comparative statics. ■

Remark (the decomposition, and the price of a moved margin). Proposition 6 reads as three separate results. It is one result, and a two-line accounting shows it. Write any transfer instrument as a pair (m_w, m_e) : what it pays while working and while not. The participation margin cannot see the pair, only the difference $\Delta = m_w - m_e$: the reservation wage is $R = s - \Delta$, or $R = s(y) - \Delta$ once the exit option's value rises with unearned cash y in the exit state (leisure normal, so $s' \geq 0$). Exactly two channels move participation. The first is $\partial R / \partial \Delta = -1$: the substitution effect, always at full strength. The second is $\partial R / \partial y = s'$: the income effect, whose size the empirical literature puts near zero (Hoynes and Rothstein, 2019). The proposition's three instruments are three values of Δ . The in-work benefit sets $\Delta = b$. The out-of-work benefit sets $\Delta = -b'$, its income term reinforcing its substitution term. The unconditional transfer is the unique instrument with $\Delta = 0$: its compensated participation response is zero as an identity, and its uncompensated response is s' -sized in fact. In EITC/NIT terms, conditional programs move the participation margin through the substitution channel by design; a universal transfer has no substitution channel to work through.

The corner prices a moved margin in output: every misplaced hour of participation costs $|c \cdot \bar{p} - s|$, in either direction. Work summoned below the outside option burns $s - c \cdot \bar{p}$ per hour (6ii); income-effect exit above it burns $c \cdot \bar{p} - s$. An instrument's output cost is the distance it moves the margin times that price: full strength for b and b' , s' -sized for u . Deadweight is a stricter ledger. b and b' appear on it because substitution drives a wedge between the work–exit price $w - s$ and the social return on the hour; hours a wedge misplaces are excess burden in the standard sense. u drives no wedge: its exit is an income effect, efficient at post-transfer valuations. Its burn is forgone output priced at the stale pre-transfer s , not a welfare triangle; zero compensated response, zero excess burden. With s' near zero, even that output term prices a near-empty set.

What survives in the linked regime is incidence, not efficiency: the little exit u induces raises the base wage through Proposition 1's comparative statics, a transfer from firms to remaining workers. Speenhamland ran that leak forward, an in-work subsidy accruing to employers through a lower wage; u runs it backward, accruing to workers through a higher one. (An insurance reading of the same decomposition — conditioning as coverage — is deferred to the appendix.)

Proposition 7 (Funding). (i) In the corner regime, labor demand is perfectly elastic at gross wage $c\bar{p}$, so a payroll or income tax t cannot pass to firms; workers' net wage is $(1-t)\cdot c\bar{p}$. With full participation the transfer is $u = t\cdot c\bar{p}$, and disposable income is

$$(1-t)\cdot c\bar{p} + t\cdot c\bar{p} = c\bar{p} \text{ — identically, for every } t, \text{ at full participation.}$$

A wage-funded universal transfer in the corner regime is a closed loop through the same pockets. With partial participation it is worse than futile: the transfer spreads over N people while the tax falls on fewer, each worker's disposable income drops below $c\bar{p}$, and when it crosses s , marginal workers exit, the base shrinks, and u falls — a self-undermining spiral. (ii) Land's supply is fixed and its rent is a residual, so a tax on site rent capitalizes into the land *price* while leaving the rent flow and land use unchanged: the base cannot contract in response to being taxed. By (6i), the transfer it funds moves no margin either. That conjunction — untaxable-away base, distortion-free transfer — is the theorem. Two practical caveats, acknowledged: assessment must separate site value from improvements — a century-old valuation literature, hard but workable — and a near-complete tax capitalizes into land prices, compressing the collateral channel during transition. Neither touches the invariance logic. (iii) By Proposition 3's corollary, the non-empty funding bases in the corner regime are physical rents, institutional rents, transition rents, and thin interest; the wage column is blank. Funding transfers from the wage base is therefore impossible; and among the non-empty columns the site-rent base is distinguished rather than merely available — institutional rents are policy-elastic by construction and self-liquidating where labor-attached (Section 4), transition rents expire, interest is thin, while the land column, by (ii), is the only one that cannot empty itself. What is proved is base-class necessity plus within-class dominance — not, strictly, instrument necessity.

Proof. (i) Incidence under perfectly elastic demand, plus arithmetic. (ii) Fixed supply plus residual rent, plus (6i). (iii) Exhaustiveness of the taxonomy. ■

Remark (the regime dial). Away from the corner, everything is continuous. With labor-demand elasticity ε_D — which Proposition 1(ii) maps to the inverse slope of $\tilde{p}$ at the margin — and participation elasticity ε_S , workers bear $\varepsilon_D/(\varepsilon_D+\varepsilon_S)$ of a payroll tax, and exactly that fraction of a wage-funded transfer is shell game; the in-work leak of Proposition 6(ii) is the same ratio mirrored. The corner is the $\varepsilon_D \rightarrow \infty$ endpoint, not a knife-edge: the theorems degrade smoothly in observable incidence. Which makes the regime measurable — every payroll-incidence estimate is, in this model, a reading of the slope of the link.

The converse deserves equal billing: **in the linked regime, standard redistribution works roughly as advertised** — labor demand is inelastic enough that taxes stick where aimed. The theory does not say the textbook was wrong; it says which regime the textbook was written in, and identifies the switch. Redistribution purely among labor, in a fully wage-linked economy at machine parity, pins its floor to subsistence no matter how the rules shuffle; the George program — tax the site rent, pay the uniform dividend — was suspended by the industrial era's steep schedule and is resumed by its flattening. George is not merely elegant; in the corner, his base is the necessary one.

7. The remedy, priced

Section 6 priced the failure; this section prices the remedy, in three steps: whether rents can fund the floor (feasibility), how long the natural floor lasts (enclosure), and what funds the crossing (the mix).

Feasibility: the coverage ratio

Necessity is not sufficiency: Proposition 7(iii) says the rent column is the one that cannot empty itself, not that it is tall enough. The stark remark of Section 5 asks whether anyone without a site eats; this subsection prices the question.

Before the algebra, the question in words: if the state captured every dollar of site rent and mailed each person an equal check, could everyone afford subsistence? The coverage ratio κ is that check divided by the cost of one person's subsistence bundle, so $\kappa \geq 1$ means yes. One feature of the answer is worth flagging in advance: the fiscal question will turn out to be a land question in disguise. The floor is fundable exactly when there are enough acres, directly and embodied in goods, for everyone's bundle.

Setup. Per person, subsistence is a bundle (g_s, h_s) — g_s units of the machine-made good, h_s of direct land services — costing $P_s = p_g \cdot g_s + r \cdot h_s$ (the bare P of Proposition 4's proof is the aggregate index; P_s is the floor's price tag). Full capture of site rent funds a universal transfer of rT/N per head; partial capture funds the proportional fraction, so the clean object is coverage at full capture, $\kappa \equiv rT/(N \cdot P_s)$, with $q \equiv r/p_g = (1-a)/(\ell \cdot \bar{p} \cdot \bar{L})$ as in Proposition 4(ii).

Proposition 8 (Feasibility of the rent-funded floor). In the corner regime: (i) the coverage ratio is

$$\kappa = q \cdot T / (N \cdot (g_s + q \cdot h_s)),$$

strictly increasing in q from 0 toward the physical land ratio $T/(N \cdot h_s)$, which it never attains: machine improvement raises coverage on every margin — along falling a only as far as the bounded $q = 1/(\ell \cdot \bar{p} \cdot \bar{L})$ allows (echoing 4(ii)); along $\bar{p} \rightarrow 0$ and $\ell \rightarrow 0$ all the way toward the ratio itself. (ii) The floor is fundable, $\kappa \geq 1$, iff $q \geq q^* = N \cdot g_s / (T - N \cdot h_s)$, a finite threshold that exists iff $T > N \cdot h_s$; if land per head falls short of subsistence's direct land content, no machine progress funds the floor. (iii) The fiscal constraint is the land constraint in disguise: $\kappa \geq 1$ holds exactly when $N \cdot (g_s/q + h_s) \leq T$ — the direct plus embodied land content of N subsistence bundles fits in the land there is. Since national income is rT identically (5ii) and the base cannot contract under taxation (7ii), the rent-funded floor is feasible whenever any allocation of the economy could deliver N subsistence bundles. The wage-funded route stays infeasible at every q (7i).

Proof. (i) Substitute P_s and divide through by p_g : $\kappa = q \cdot T / (N \cdot (g_s + q \cdot h_s))$; $d\kappa/dq = T \cdot g_s / (N \cdot (g_s + q \cdot h_s)^2) > 0$, with limits 0 and $T/(N \cdot h_s)$, and the margins move q by Proposition 4(ii). (ii) Set $\kappa = 1$ and solve for q . (iii) One unit of the good carries $\ell \cdot \bar{p} \cdot \bar{L} / (1-a) = 1/q$ units of land through the recursion, so N bundles demand $N \cdot (g_s/q + h_s)$ of the T available; rearranging $\kappa \geq 1$ gives exactly this inequality. ■

The demolition funds its own remedy: the same q that empties the wage column (4ii–iii) runs the coverage ratio up its curve, so George's program grows more fundable precisely as it grows more

necessary. The failure modes are exhaustive and physical, not fiscal: too little land per person ($T \leq N \cdot h_s$), or progress not yet far enough ($q < q^*$). A worked instance ($a = 0.6$, $\ell = 0.2$, $\bar{p} = 1.8$, $\bar{L} = 1$, so $q = 10/9$; $T = 100$; unit bundle $g_s = h_s = 1$): fifty people are covered with a nineteenth to spare ($\kappa = 20/19$); eighty must wait for q to reach 4; one hundred twenty — land per head below the bundle's land content — are never covered. The ratio says which world you are in.

κ is measurable — Section 8 measures it (Figure 6) — and the floor's own spending feeds the base back through the closure: favorable in direction, a tendency in magnitude, priced in the appendix.

The commons, priced

Every floor in this paper rests on the outside option s . The wage cannot fall below it (Proposition 1(iii)), and Proposition 8's bundle is priced to replace it. Section 2 noted in passing that s quietly presupposes access to some land. That debt comes due here. Model the exit life directly. A worker who leaves the market produces for herself: the exit yields an autarkic keep s_o and requires h_e of land services, quality-adjusted like the bundle's h_s , so no particular parcel is needed. The keep uses no machines by assumption; the remark's caveat (d) relaxes it. Autarkic describes production, not exchange: the plot's rent is a market price. If no plot is affordable, exit degrades to a dependency floor $s_d < s_o$: the value of living on someone else's land, at someone else's sufferance. Values are in the goods numeraire of Proposition 4, with $q = r/p_g$ the goods price of land services.

Proposition 9 (The enclosure margin). This proposition prices the outside option itself: when the exit plot is free, what it costs once no land lies idle, and whether the natural floor survives until a funded one arrives. (i) While land demand falls short of the stock T , the marginal parcel rents at its reservation value, zero. The commons is the idle margin of the land market. The exit plot is therefore free, and $s = s_o$. This is why s could stand as an exogenous parameter through Propositions 1–8: land was not yet scarce at the margin. (ii) The idle margin closes when land demand reaches the stock; the corner guarantees this, since land clearing $T_P + T_H = T$ (5i) leaves no parcel unemployed. Once it closes, the exit plot pays the ruling rent and

$$s(q) = \max(s_o - q \cdot h_e, s_d),$$

falling one-for-one with $q \cdot h_e$ and reaching the dependency floor at $q_{enc} = (s_o - s_d)/h_e$. Finite machine progress completes the enclosure: the margins that demolish the wage (Propositions 4(ii)–(iii)) also demolish the exit from it. (iii) The race. By (8i) the coverage ratio $\kappa(q)$ rises with q ; by (ii) the exit value $s(q)$ falls. The natural floor dies at q_{enc} ; the fundable floor arrives at q^* (8ii). The floor is fundable before the commons dies iff the population is small relative to the land:

$$q_{enc} \geq q^* \text{ iff } N \leq q_{enc} \cdot T / (g_s + q_{enc} \cdot h_s).$$

Otherwise a gap opens: commons gone, floor not yet fundable. In the gap, the next subsection's instruments are rescue, not optimization.

Proof. (i) With demand short of the stock, some parcel of reservation rent zero sits idle; any positive rent on the marginal parcel is undercut by an idle neighbor. Exit costs nothing. (ii) Once demand reaches the stock, no idle neighbor remains; at the corner (5i) makes this an identity, every parcel employed in production or housing. An exit plot is a housing use and pays the same q per unit as any other. The keep is $s_0 - q \cdot h_e$ until s_d binds; equating the two gives q_{enc} . (iii) Substitute $q^* = N \cdot g_s / (T - N \cdot h_s)$ from (8ii) into $q_{enc} \geq q^*$ and rearrange; $T - N \cdot h_s$ is positive by Proposition 8's fundability condition. ■

Remark (desperate supply, and the floor that walks back in). Three consequences and a caveat. (a) By (ii) the trigger for enclosure is land scarcity, not machine parity: the idle margin can close while the wage schedule still has slope. There a falling s pushes the participation margin inward. Workers who would have exited now stay, at wages they would have refused. More participants mean a lower wage, by Proposition 1's comparative statics. (At the corner the wage is pinned; enclosure moves only the exit value.) Enclosure manufactures labor supply. This is the model's reading of the historical enclosures, in Section 3's pre-industrial configuration: land fully claimed, schedules still sloped. Not a land-market event with labor-market side effects: one event posting to two ledgers. (b) The cancellation in 6(i) survives any $s(q)$: u enters both sides of the participation comparison, so it still moves no margin. Its meaning shifts. Below q_{enc} , the u that holds the floor at s_0 bills exactly the $h_e \cdot q$ the enclosure took. Past q_{enc} the household pays no rent and the guaranteed floor is $s_d + u$: the transfer replaces the commons rather than supplementing it. The take is then capped at $s_0 - s_d = h_e \cdot q_{enc}$; restoring s_0 bills that sum, not the still-rising plot rent $h_e \cdot q$. (c) The funding circle. The exit plots are housing land, inside T_H , so their rents are already inside rT . The replacement bundle carries a plot of its own: h_s sits in P_s . So the flow that priced the commons away is the same flow that funds its replacement. (d) One caveat: holding s_0 constant treats exit as autarkic. A machine-tooled exit adds a falling goods term to the exit bundle, and the conclusion stands wherever that bundle keeps a positive land share σ — the logic of Proposition 4(iii), applied to the woods.

The mix on the way down

Proposition 8 fixed the destination: coverage κ — site rents over the cost of the floor, $rT/(N \cdot P_s)$ — climbs toward the feasibility threshold as automation advances. Proposition 9 named the danger: enclosure can close the commons before coverage arrives; in that gap the floor must be funded now. Until κ reaches one, a full floor needs revenue beyond site rents. The only other bases still standing are the transition's own: dynamic rents from the buildout, institutional rents not yet liquidated, and the wage flow, not yet zero and still financing consumption. Two instruments reach them, and each is blind where the other sees. A land-value tax (LVT) reaches every site rent, consumed or saved, whoever owns it; it moves no margin (7ii); it needs a cadastre; and it cannot touch a bottleneck profit or a royalty, which attach to no site. A uniform VAT reaches whatever finances consumption — including exactly those non-site rents — and needs no valuation; but the slice financed by wages makes it, to that extent, a payroll tax in disguise. Write E for aggregate consumption spending and λ_C for its wage-financed share; measured below, λ_C stands at 0.72 and is falling, κ at 0.33 and rising.

Proposition 10 (The mix frontier). Fund the floor with both instruments, choosing rates to minimize deadweight loss. Let a land tax at rate $t_L \leq 1$ on the rent flow rT ($t_L = 1$ is full capture) and a uniform VAT at rate t_V on E together raise the floor $N \cdot P_s$. Assume the VAT's deadweight is the standard half-square on its wage-financed slice, $(\psi/2) \cdot \lambda_C \cdot E \cdot t_V^2$, with ψ scaling the compensated participation response (the regime dial's elasticity ε_S). (i) If $\kappa \geq 1$, the optimum is the land tax alone: $t_V = 0$, zero deadweight. (ii) If $\kappa < 1$ — the measured present — the optimum fills the rent base first: $t_L = 1$ and $t_V = N \cdot P_s \cdot (1 - \kappa) / E$. The minimized deadweight is proportional to $\lambda_C \cdot (1 - \kappa)^2$ at a given floor-to-consumption ratio. (iii) At the corner the bases merge: by Proposition 5's identity, all consumption spending resolves into rent income, so a uniform VAT and a full land-value tax become one tax, and the mix question retires.

Proof. The land tax's marginal deadweight is zero at any rate up to one (7ii). The VAT's is positive and rising, because only its wage-financed slice moves a margin; rent-financed consumption is inframarginal. Cheap revenue precedes costly revenue: the rent base fills first, and $t_L = 1$ whenever $\kappa < 1$. The residual requirement is $N \cdot P_s - rT = N \cdot P_s \cdot (1 - \kappa)$, so $t_V = N \cdot P_s \cdot (1 - \kappa) / E$. Substituting t_V into the half-square hypothesis gives

$$D^* = (\psi/2) \cdot \lambda_C \cdot (N \cdot P_s)^2 \cdot (1 - \kappa)^2 / E,$$

which is proportional to $\lambda_C \cdot (1 - \kappa)^2$ at a given floor-to-consumption ratio $N \cdot P_s / E$. For (iii): at the corner, Proposition 5 makes all consumption spending rent income, $E = rT$, so a uniform VAT and a full land tax draw on the same base. ■

Remark (what the VAT is for, and what the crossing costs). On the way down, the VAT's assignment is the non-site rents: the buildout's bottleneck profits, the licensed and patented margins that no cadastre can see. That reach decays by construction. Dynamic rents expire when their bottlenecks clear; labor-attached institutional rents self-liquidate with the demolition (Section 4). Proposition 10(ii) prices the crossing: the cost of funding the floor is the product of two measured series, the wage-linkage of consumption and the squared shortfall of coverage, and both are moving the right way. At today's pair (0.72, 0.33) the index $\lambda_C \cdot (1 - \kappa)^2$ stands at 0.32; at, say, (0.50, 0.60) it falls to 0.08. The weights travel with the regime: the VAT is sized to the coverage shortfall and shrinks as κ climbs; the land tax stands at one throughout. At the margin the land tax strictly dominates — zero deadweight against positive — so the optimum is a corner in t_L , not a compromise. But neither base contains the other: the land tax cannot reach a bottleneck profit, the VAT cannot reach a saved rent. Both carry positive weight for the whole crossing, and neither retires before the bases merge. The VAT is scaffolding — it bears weight only while the structure that replaces it goes up.

8. Measurements: the United States, 1962–2025

How wage-linked is the economy the demolition is aimed at? Working from BEA national accounts (annual NIPA series via FRED), every contestable classification — the labor share of proprietors' income, the progressivity tilt of personal taxes, corporate incidence, transfer funding rules, saving-attribution order, deficit attribution — was computed under all defensible variants; figures below are medians with bands across the full rule grid in brackets. None of this tests the

propositions; it measures the exposures that make them consequential — the model's test surface is Section 9.

Object	Value (year)	Note
Wage-linked share of consumption financing, λ_C	0.72 [0.66–0.81] (2025)	down from ≈ 0.80 in 1965
Wage-linked share of tax revenue, λ_R	0.68 (2025)	0.64 in 1965, ≈ 0.70 by the mid-1980s, flat since
Work- or means-conditioned share of benefits	≥ 0.84 floor; ≈ 0.99 by withdrawal rule (2024)	the only sizable unconditional cash transfer is Alaska's oil dividend
Government purchases paid as direct public wages	0.47 (2023)	$\lambda = 1$ exactly; the rest is procurement and capital consumption
Transfers, share of GDP	15.8% (2025)	borrowing funds 19% of transfers in 2025
Household net worth / GDP	5.8 (2025)	3.6 in 1965, 3.4 in 1980, 6.1 in 2021
Coverage ratio of the rent-funded floor, κ (Proposition 8)	0.33 [0.18–0.59] (2025)	≈ 0.05 in the 1950s; full-capture site rent vs. poverty-line floor; land measures are lower bounds

Two structural findings sit underneath the table. First, **the corporate veil**: the labor share of GDP is in the mid-fifties, yet the labor share of income households actually receive is 73% — because of what happens to the \$9.8T gross operating surplus (2023): \$3.8T is consumed by depreciation, \$0.85T retained by business, and \$0.6T intercepted by corporate tax before any of it becomes spendable income. Depreciation, the largest term, is Proposition 3 wearing accounting clothes: most of gross capital income is the machines maintaining themselves — the a-c term of the recursion, visible in the national accounts. Second, **the scissors in time**: the economy's wage-linkage (λ_C) has drifted down for sixty years while the fiscal system's wage-exposure (λ_R) rose and locked — the state deepened its dependence on exactly the base the technology erodes. The scissors' residual is the borrowed channel of Figure 5: promises whose ultimate payer is undecided, and which must eventually harden into claims on whichever column the regime leaves standing — wage taxes if the link holds, rents or inflation if it does not.

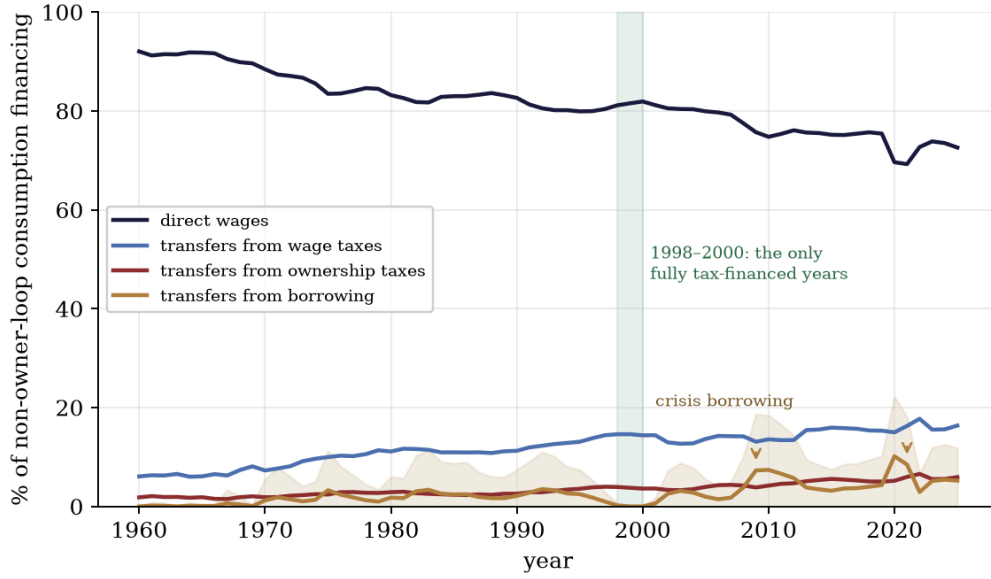


Figure 5. Financing of U.S. consumption outside the owner loop — the owner loop being the slice financed directly by household capital income, owners consuming their own returns — 1962–2025 (medians across the classification-rule grid; shaded band spans deficit-attribution rules). Direct wages' share of this narrower object — not the λ_C of the table — has ceded twenty points in sixty years, in a ratchet: every recession bites, every recovery returns only part. Transfers grew on a funding mix that stayed roughly quarter-ownership; the borrowed channel reached 19% of transfers by 2025, and only in 1998–2000 did it vanish. Funding attribution is pro-rata accounting, not estimated incidence.

The remedy, measured. The coverage ratio of Proposition 8 is a measurable object, and Figure 6 measures it: the numerator is the aggregate site-rent flow the national accounts carry in pieces — Z.1 land residuals (market-value real estate minus replacement-cost structures, household and business) under capitalization-rate variants, with the housing-services flow as a lower-bound member — and the denominator is population times a costed subsistence bundle, priced by the Census thresholds' own CPI-updated logic. Every contestable step goes through the same rule grid as the table: all defensible variants, medians with min–max bands. The result: $\kappa = 0.33$ [0.18–0.59] in 2025, roughly 0.05 in the 1950s. The ratio has climbed for seventy years, as Proposition 8 says it should — the demolition funding its own remedy — and the $\kappa = 1$ line is not yet in reach. Every land measure used is a lower bound, so the truth likely sits above the band's middle.

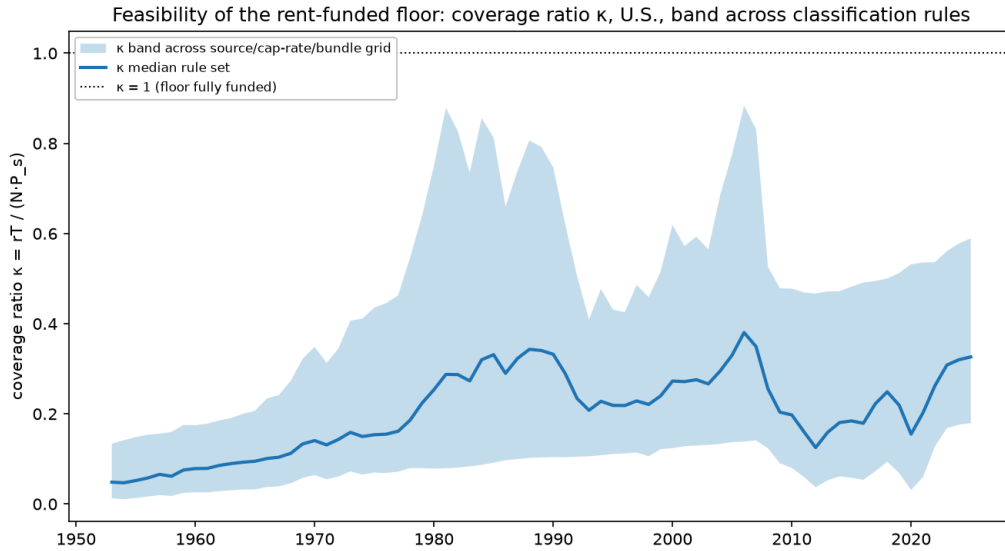


Figure 6. Coverage ratio $\kappa = rT/(N \cdot P_s)$ of the rent-funded floor, U.S., 1953–2025: full-capture site rent against a universal subsistence floor, median and min–max band across the land-source $\times$ capitalization-rate $\times$ bundle grid. The band's width is substantially the capitalization-rate spread — the 1980s bulge is the rate cycle, not land — and the Z.1-residual members carry the known post-1995 caveat; business-structure coverage ends in 2020.

The fork, drawn. Prediction 8's mechanism is visible in the rawest series available: the same average paycheck, indexed at 1964, buys nearly four times the durables and a fifth less shelter — a near-fivefold fork between the machine-made and the land-priced denominators, opening exactly as machine capability spread (Figure 7). The divergence is Proposition 4's bookkeeping in public data: gadgets follow γ_M , ground follows r .

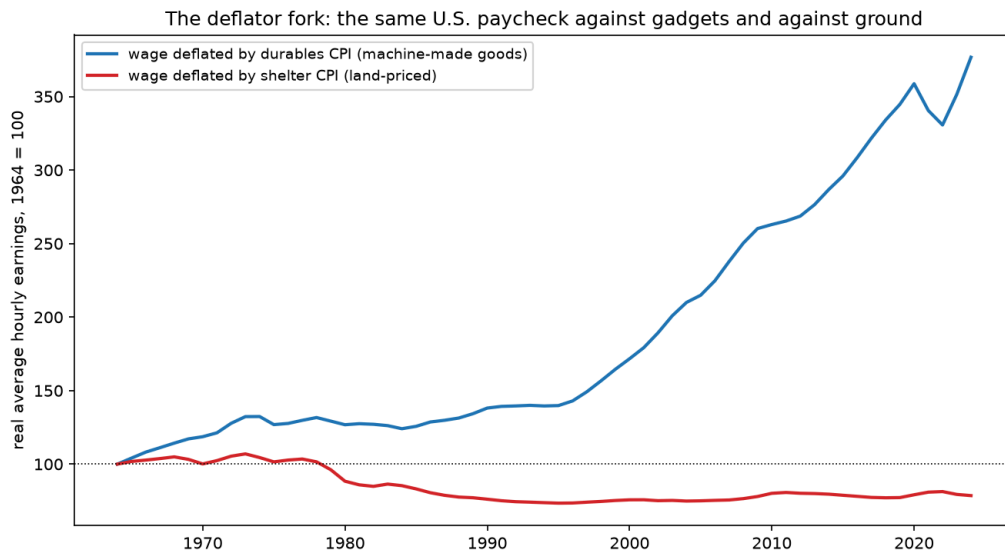


Figure 7. The deflator fork: U.S. average hourly earnings (production and nonsupervisory) deflated by the durables CPI and by the shelter CPI, 1964 = 100, complete calendar years through 2024. The fork ratio reaches 4.8 $\times$.

9. Predictions

Where the model can be wrong, in thirteen entries; the bracketed tags name each one's test.

1. **Within-group compression precedes between-group collapse.** Groups exposed to automation lose most at the 70th–95th within-group percentiles; dispersion inside exposed groups falls even as gaps between groups widen. [By construction for 1980–2016 — the mechanism was transplanted from this finding (Acemoglu–Restrepo 2026); the predictive content is its persistence, and its inversion where protection holds the adoption decision (Prediction 2).]
2. **Protection accelerates replacement — where it prices labor.** At equal capability, the unionized, licensed, or premium-priced version of a task is automated earlier than the unprotected version; where protection instead holds the adoption decision — law, contract, scope rules — the ordering inverts, adjustment turns punctuated, and protection converts toward legal form under pressure, with a rising share of collective agreements carrying automation clauses as the direct trace. Price-based human-provenance premia get automated away or normalized away; only legal task reservations endure. [Cross-regime contrast: testable with matched task/occupation data crossed with contestability.]
3. **The transition can be negative-sum.** Periods of intense wedge demolition show automation raising output while yielding little or no net TFP gain. [Estimated for 1980–2016: rent dissipation offset 60–90% of automation's productivity gains.]
4. **Wages pile up toward the floor, so traps worsen without rule changes.** The share of workers inside benefit-withdrawal bands — facing high effective marginal tax rates — rises mechanically as compression proceeds. [Testable in effective-marginal-tax-rate microdata.]
5. **The labor share's decline is not mean-reverting.** It resumes after each cyclical recovery rather than reverting to the postwar mean, deepening with each wave of capability uniformity. [Ongoing; the decline is global — Karabarbounis–Neiman 2014.]
6. **The rising capital share is land in disguise.** Measured capital income increasingly resolves into housing and site value rather than returns on reproducible equipment. [Consistent: Rognlie 2015, Knoll–Schularick–Steger 2017; continued concentration predicted.]
7. **AI value attaches to sites, power, and water — not weights.** Model-only firms' margins compress through distillation and copying; the durable balance-sheet items are datacenters, land, energy contracts. [Unfolding.]
8. **Real wages fork by deflator.** Wage series deflated by machine-producible goods hold or rise while the same series deflated by shelter-heavy, land-content bundles stagnate or fall, and the gap widens with automation; the land-content share of low-income budgets climbs. The divergence is the mechanism of Proposition 4, not a measurement nuisance. [Drawn in Figure 7: a near-fivefold fork since 1964; sharpens as parity spreads.]
9. **The fiscal scissors widen.** Wage-linked revenue exposure stays high while the wage-linkage of the economy falls; crises ratchet transfer growth financed by borrowing, and the borrowed channel's ultimate incidence stays undecided until the regime decides it. Operationally: each recession lifts the borrowed share of transfer financing by several points, with less than full retracement. [Visible in Figure 5.]
10. **Payroll incidence migrates onto workers.** As schedules flatten, the estimated pass-through of payroll taxes to net wages rises toward one, and wage-funded transfer expansions show progressively fuller worker incidence — the regime dial of Section 6, read from the incidence literature. [Testable against payroll-incidence estimates over time.]

11. **Conditional programs' failure mode migrates.** In-work subsidies shift from leaking into lower wages (linked regime) toward financing negative-value make-work (corner regime) as machine parity crosses outside options occupation by occupation. [Testable as parity thresholds cross.]
12. **The rescue is hollow where it operates.** Where human-essential expenditure shares rise (Baumol concentration), aggregate labor share and median wage diverge; within the human-essential sectors, income concentrates with reach (the top share rising with the broadcastable fraction); co-present and legally reserved premia hold while certified-at-a-distance premia erode toward their fraud bound. [Machinery in Section 10. Testable: sectoral labor shares against within-sector top-income concentration; provenance-premium panels as verification technology shifts.]
13. **The exit option is priced by rents.** Where the land content of subsistence rises, exit substitutes toward dependency: young-adult coresidence, multigenerational households, and household-formation delays track rent-to-wage ratios, not unemployment alone — and track them less where commons or cheap marginal land remains. [Testable in household-formation panels against local rent indices.]

10. Assumptions, honestly

Where a skeptic should push, in descending order of importance.

Fixed task space — the real dichotomy: Acemoglu and Restrepo (2018) derive reinstatement: falling wages redirect invention toward new labor tasks, and for two centuries that counterweight operated. This paper closes that margin deliberately, and the closure should be read as the pessimistic branch of a stated fork. Branch one: reinstatement continues — which requires a standing reservoir of latent tasks where humans retain comparative advantage after each advance; that reservoir is the link, so the stabilizer presupposes what is at issue. Branch two: the reservoir thins — the branch supported by the displacement-versus-reinstatement accounting since 1987 and by the generality of AI, which targets the task-creation process along with the tasks. On branch one this paper's corner is postponed and its propositions describe a direction of travel; on branch two they describe a destination. The paper takes branch two and says so.

Exogenous wedges: wedge locations and enforcement mode are taken as given; if institutions regenerate wedges or convert them to quantity form, stage one slows and the fortified stratum's drag compounds — the tank refills at one end while it drains at the other (the fortification remark tracks this; whether it binds is political economy, outside this model's machinery).

Rationed wedge jobs: the participation margin is evaluated at the base wage (queue logic); a lottery convention softens the floor by the mean wedge and flips nothing downstream.

Checklist (Leontief-over-tasks) production, one good, homogeneous workers: CES aggregation softens thresholds into bands; heterogeneity makes every statement type-by-type; regimes and propositions survive.

Human-premium tasks: a theorem rather than a judgment — Aghion, Jones, and Jones (2019) supply the mechanism and Proposition 11 prices it: the rescue requires a human-essential task set both non-automatable and non-substitutable; preference-membership binds only where humanity can be verified (co-presence or provenance law); and where the rescue operates it is

distributionally hollow. What stays a judgment is the premise — how much capability residue persists: the first entry's fork, unchanged.

Transition dynamics unmodeled (and the border, one page): Section 11 opens the economy — trade as the early waterline, rents to the producing jurisdiction's land, the border-split bases, migration flagged. The buildout era's very real bottleneck rents are classified (dynamic scarcity) but not tracked.

The strongest objection, as a theorem

Suppose some tasks never fall. Let $K \subset [0,1]$ be the set of tasks machines cannot hold, with measure $k > 0$ and base wage w_K . Membership may have any source: a capability residue machines never close, a preference for human hands that buyers enforce, or a law that reserves the work. Outside K the per-task machine cost is $c/\gamma_M(x)$; take it uniform at m , collapsing along Section 5's margins. Aghion, Jones, and Jones (2019) raise the strongest objection this paper faces: if a human-essential set persists, Baumol dynamics concentrate expenditure on it, and labor's share can survive automation indefinitely. The objection deserves a theorem rather than a rebuttal. The proposition below grants the mechanism and prices the premise; Baumol's cost disease will be derived from the task structure, not assumed.

Proposition 11 (The Baumol fork). Three limits organize the answer: costs, expenditure, and the division of income. Extend the checklist economy of Section 2 to a family of goods, each its own checklist of tasks, with Leontief production, human productivity on K uniform at γ_L , and $m \rightarrow 0$ outside K .

(i) *Cost concentration.* Any good whose checklist contains the whole of K has unit cost $C = k \cdot w_K / \gamma_L + (1-k) \cdot m \rightarrow k \cdot w_K / \gamma_L$ as $m \rightarrow 0$, so labor's share of that good's cost tends to 1.

(ii) *The fork.* If households substitute between K -content and K -light substitutes with elasticity η , then as machine prices collapse the expenditure share of K -content tends to 1 if $\eta < 1$, to the taste weight θ if $\eta = 1$, and to 0 if $\eta > 1$. The rescue forks on substitutability, not on sentiment.

(iii) *The terminal split, on the $\eta < 1$ branch.* Give households CES demand with common elasticity $\eta < 1$ over three categories: the machine-made good at price p_g , direct land services at rental r , and K -labor services, with taste weights θ_g , θ_T , θ_K . Then terminal expenditure concentrates on the two non-collapsing categories: the machine-made share tends to zero, and

terminal share of land services $= \theta_T r^{1-\eta} / (\theta_T r^{1-\eta} + \theta_K w_K^{1-\eta})$, the rest to K -labor.

Call K -labor's part of this split the *K-share*. Terminal labor income is K -labor income, so labor's aggregate share tends to the K -share; surviving non- K wages only add, making the K -share a lower bound. The corner regime of Propositions 3–8 is exactly the $K = \emptyset$ boundary of this theorem: the paper's whole baseline is one edge of the present result.

Proof. (i) The K -tasks cost $k \cdot w_K / \gamma_L$ however cheap machines get and the rest cost $(1-k) \cdot m$; the second term vanishes as $m \rightarrow 0$, so labor's cost share $(k \cdot w_K / \gamma_L) / C \rightarrow 1$. (ii) Under CES demand the dearer good's expenditure share rises with its relative price iff $\eta < 1$, is flat at $\eta = 1$, and falls iff $\eta > 1$; by (i) the relative price of K -content diverges as $m \rightarrow 0$, forcing the three limits. (iii) Each category's share is its taste weight times its price to the power $1-\eta$, normalized by the sum; p_g collapses with m , so for $\eta < 1$ the term $\theta_g \cdot p_g^{1-\eta}$

vanishes while the r and w_K terms stand, leaving the displayed split. Setting $\theta_K = 0$ (K empty) sends the land share to 1 and returns Propositions 3–8 unchanged. ■

Remark (credence: the provenance problem). A K -membership that rests on preference binds only if provenance can be verified. Suppose buyers carry a latent taste φ_H for verified human work, verification catches a fake with probability v , and a caught faker pays a legal penalty f for provenance fraud. The sustainable premium is $\pi_H \leq \min(\varphi_H, v \cdot f / (1-v))$. The second argument is a no-entry condition: a faker passes with probability $1-v$ and pays f otherwise, so any premium above $v \cdot f / (1-v)$ makes fraud profitable and draws entry until the certificate is worthless. With price-only enforcement, so that $f \rightarrow 0$, the premium collapses for every $v < 1$. Certified-human markets unravel through fakes, not robots: Proposition 2's self-liquidation of price-mediated protection by another door. Two anchors survive the arithmetic. Provenance law sets $f > 0$, converting the preference into quantity exclusion: the fortified set of Section 4. Co-presence drives $v \rightarrow 1$: humanity verified by being in the room, where faking is impossible, not merely punished. K 's terminal core is what law reserves plus what bodies witness.

Remark (superstars: reach and the room). Where K rescues labor's aggregate share, it rescues neither the median K -worker nor the median worker. Within K , reach (recording, broadcast, replication) is machine-side. Reach serves the β -fraction of K -demand that does not require the room; following Rosen (1981), let that whole slice go to the single best performer. The co-present remainder is rationed by human capacity. With n K -workers, the top performer's share of K -income is $\beta + (1-\beta)/n$. It rises in β at rate $1 - 1/n$, and reach is what raises β . The median K -income falls to $(1-\beta)$ of the mean; that is the median inside K . The economy-wide median is lower still: wherever K employs a minority, the median worker holds no K -task and earns the corner floor. On the $\eta < 1$ branch the aggregate labor share can persist, even approach one, while the median wage still collapses to the floor. Aggregate rescue, median demolition.

The fork sends the fiscal ledger one bill and one credit. On the K branch, the subsistence bundle itself carries a K -service component: care and teaching are the human hour the floor must buy. That component does not collapse with machine prices, so P_s gains a non-collapsing term and the coverage ratio κ of Proposition 8 sits lower at every q . The credit: K -labor income is now a positive income share, so the wage column of Proposition 7(iii) is no longer blank; where K is law-made, the fortification remark prices that wage as institutional rent in wage form, taxable as rent. What survives the machines makes surviving them dearer.

11. The open economy, in one page

The model has run closed; one page opens it. The opening rests on one conceptual move, stated before the algebra performs it: the foreign wage will enter the model as a second machine rental — a rental price on human capability itself. Nothing about foreignness does any further work; all the allocation ever sees is the price. A task that leaves for a machine and a task that leaves for Shenzhen are the same event on the capability schedule at two different prices, which is why the empirics of automation and offshoring are so hard to separate. Let a tradable subset of tasks be performable abroad at foreign wage w_F , with effective foreign schedules built exactly as at home (Lemma 1 with foreign subscripts), and let machines rent at c everywhere: Proposition 3's recursion respects no border.

Proposition 12 (Trade as the early waterline). (i) A tradable task is held at home iff $w \leq \min(c \cdot \tilde{p}(x), w_f \tilde{p}_f(x))$, where $\tilde{p}_f$ deflates the home-over-foreign human edge by the same wedge. The foreign wage enters as a second machine rental — a rental on human capability. On the tradable set, the demolition of Section 4 therefore runs with a weakly higher waterline: same targeting (wedges are negative altitude in both waters), same strata, earlier timing. (ii) Offshoring is a way-station, not a destination: the parity that empties the home tradable set empties the foreign one — when machine cost undercuts w_f , the offshore task flips onward to machines wherever they sit — and the terminal allocation is the closed model's corner, with production landing where sites, energy, and water are cheapest. (iii) Rents accrue to the land of the producing jurisdiction. Through the recursion, a consuming country's spending therefore resolves partly into foreign site rents: the embodied land content of its imports. Those rents lie beyond the reach of the consuming country's LVT. A destination VAT reaches exactly that slice. The LVT reaches domestic sites whoever owns them. The border splits the bases, each instrument covering the other's cross-border blind spot. The mix frontier of Proposition 10 is interior in the open economy for a second reason.

Proof. (i) Three suppliers per task; divide the cost comparison by $\mu(x)$ as in Lemma 1. (ii) Lemma 1 with foreign subscripts. (iii) Proposition 5's pass-through runs where the machines run; the destination principle and in-rem incidence are definitional. ■

Remark (borders, rooms, and feet). Three notes and a coda. The wedge story travels: on the tradable set the best-protected jobs flood first whichever water rises — in this model automation and offshoring are one waterline read at two heights. The refuge doubles: the non-tradable set is largely the co-present set, so the room that verifies humanity (the remark to Proposition 11) is also the room the container ship cannot enter — co-presence shields twice, and the terminal K of Section 10 is unchanged by opening the border. And feet move: a jurisdiction's dividend is its own rT/N , so dividend differentials invite migration and the crowding threshold of Proposition 9(iii) is per-jurisdiction — flagged, not modeled, alongside the buildout dynamics it belongs with. For a consuming country in deficit, part of the floor's price tag is foreign land — beyond the domestic LVT, within the destination VAT. On open borders, Section 7's scaffolding hardens into structure: for a deficit consumer the VAT is not optional but constitutive.

12. What the planner would do

The preceding sections price the demolition, the migration, the floor, the crossing, and the objection; this one asks what a planner holding the model's own instruments should do, and finds the answer has been assembling itself all along.

Proposition 13 (The corner welfare theorem). The proposition says that at the corner the land-tax-and-dividend pair does the whole job of policy: it leaves private choices efficient, it can deliver any division of the rent along the way from inherited to equal, and — up to the corner's own merging of consumption and rent bases (10iii) — nothing else in the model's instrument set can do either. Take the corner regime with land fully enclosed (Proposition 9), and let the George pair be available: an LVT at rate $t_L \leq 1$ funding a uniform transfer $u = t_L \cdot rT/N$. (i) *Efficiency.* Private choices are socially right under u : the pair is production- and participation-efficient at every rate t_L . The automation margin is undistorted. The private participation rule (work iff $c \cdot \bar{p} \geq s$) coincides with the social one. The reason is that at parity the wage is the machine's replacement cost, and the machine's replacement cost is the marginal task's social value. (ii) *Implementation.* The pair spans, at zero deadweight, the one-parameter family of divisions of rT from the inherited ($t_L = 0$) to the equal ($t_L = 1$). The base cannot contract (7ii). The transfer moves no margin (6i). Nothing else in the corner is left to distort. Land rent is the economy's one real lump sum. What the second welfare theorem promises and wage taxation cannot deliver — redistribution without deadweight — the pair delivers along this family; the rest of the frontier would need person-specific lump sums, which no tax system has ever had. (iii) *Converse.* The pair is unique within the model's instrument set, up to the corner equivalence of consumption and rent bases (10iii). Work-conditioned floors violate (i): every hour of participation they misplace burns $|c \cdot \bar{p} - s|$. Wage-funded floors violate (ii) by the shell game (7i).

Proof. (i) The adoption margin compares costs the planner also faces; under u the displayed participation rule is 6(i)'s cancellation read at the social wage $c \cdot \bar{p}$. (ii) Person i with land share ω_i receives $(1 - t_L) \cdot \omega_i \cdot rT + t_L \cdot rT/N$; the family sums to rT for every t_L , and no term moves a margin by 7(ii) and 6(i). (iii) is the conjunction of 6(ii)–(iii), 7(i), and 10(iii). ■

Remark (the moral, stated plainly). Hold the technology fixed at the corner — same q , same rT — and vary only the fiscal architecture. Under the inherited, wage-linked state the base is the collapsing column: revenue follows the wage to zero, work-conditioned floors lapse with the work, and the terminal income of the claimless is s_d . Under the George pair the same economy pays everyone rT/N above the dependency floor, participation is efficient, and the machines work for whoever holds the dividend. **The corner is not a forecast of misery; it is a fork between two bookkeeping systems.** The difference between the corner as dystopia and as inheritance is a tax base — which is to say, a choice. Externality pricing composes freely on top: the pair is lump-sum-like, so a Pigouvian charge on land abuse or a carbon levy interacts with nothing here, and is priced separately, as it should be.

Remark (second best, on the way). Away from the corner the planner is not free, on three counts. The wage still carries λ_C of consumption. The wedge stock makes some privately profitable adoption socially negative: the social margin compares machine cost to the base wage, not the wedge-inflated one. The band between those two wages, of width $(\mu - 1) \cdot w / \gamma_L$ per task, is negative-sum adoption — the fuel of Section 4's transition loss. And the fortified stratum blocks even efficient adoption. Production efficiency survives the second best (Diamond and Mirrlees, 1971): tax rents and consumption, never the adoption margin itself — the fortified equilibrium is precisely the violation. The second-best schedule is then the measured schedule. Fill the rent base first. Bridge with the VAT at deadweight $\lambda_C \cdot (1 - \kappa)^2$ (10ii). Let the weights travel as both

series move. And treat the gap of Proposition 9(iii) — commons dead, floor not yet fundable — as the emergency it is. The planner's timetable is not a doctrine; it is two time series the national accounts already publish.

Appendix: deferred remarks

Remark (the manager, limiting case). The manager is wedge-holder and adoption-decider in one chair, disciplined only by entry and takeover. That discipline is priced, so the shelter erodes rather than fortifies — slowest where product markets are most sheltered. The logic is the strike threat's: entry and takeover price the shelter at the cost of replacing the sheltered, and automation feasibility is precisely a fall in that cost.

Remark (substitution: the CES dial). Replace Cobb–Douglas with CES elasticity η between goods and land services, and write $q \equiv r/p_g$. The land share of expenditure becomes

$$s_h(q) = \sigma \cdot q^{1-\eta} / (\sigma \cdot q^{1-\eta} + 1 - \sigma),$$

and as machine improvement sends $q \rightarrow \infty$ (Proposition 4(ii)): $s_h \rightarrow 1$ for $\eta < 1$ — complements; the squeeze sharpens and housing eats the budget — while $s_h \rightarrow 0$ for $\eta > 1$ and substitution bounds it. Prediction 8 is thereby promoted from a deflator observation to an estimate of η ; the climbing shelter share of low-income budgets is the $\eta < 1$ signature.

Remark (heterogeneous land and the idle margin). With heterogeneous parcels the aggregates of Proposition 5 are unchanged: rT reads $\int r(\tau) \cdot d\tau$ over parcels in use, parcels at the idle margin earn their reservation rent of zero, and r in the identities is the rent schedule, not a single number. Nothing in the proposition requires every acre to be employed — only that spending, wherever it goes, crosses a site.

Remark (corner-above). With labor still present at machine parity the identity weakens to: *all non-wage income is site rent*, $GDP = N \cdot c \cdot \bar{p} + rT$, with the wage bill pinned at parity and vanishing along the $\bar{p} \rightarrow 0$ margin, recovering Proposition 5's identity in the limit. The full mixed-case accounting adds notation without changing where anything goes.

Remark (insurance: conditioning as coverage). The states an instrument pays in are the states it insures. A work-conditioned claim like b pays only in the working state, precisely the state the collapse removes; the policy lapses when the insured event arrives. An exit-conditioned claim like b' does pay in the loss state, but it manufactures entries into that state and taxes the first hour back (6iii). The unconditional transfer is written on no state, so its guarantee survives the regime change that voids the one claim and poisons the other; in the corner it is pure insurance at zero deadweight. Proposition 7(ii) is the same property on the base side: the surviving funding source is the one written on no margin the collapse can move. The contract that pays in every state is the one written on none.

Remark (the base feeds back — tendency, not theorem). The floor of Proposition 8 is spent partly on land services; in the corner that spending resolves into rents (Proposition 5), lifting r and with it the base rT . The direction is favorable — the loop closes the right way — but the magnitude rides on distribution and substitution; claimed as tendency only.

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Data and computation. All U.S. figures are gross annual flows computed from BEA national accounts (annual NIPA series retrieved via FRED, 2026 vintage), with Federal Reserve Z.1 balance-sheet lines, BEA fixed-assets stocks, and the CPI added for the feasibility coverage ratio. Series windows follow their sources: the financing splits run 1962–2025, the coverage ratio 1953–2025, the deflator fork 1964–2024 on complete calendar years. Revenue is decomposed into wage- and ownership-linked parts (payroll to wages; personal income tax split by the labor/capital mix of household income with a progressivity tilt; corporate and sales taxes bracketed between standard incidence readings; property taxes to ownership); transfers draw on the pool pro rata (accounting attribution, not incidence estimation), with the borrowed share equal to the government current-account gap times transfers' share of current expenditure, attributed under three labeled rules. Benefit conditionality follows program rules: conditioned on work history, or withdrawn as earnings rise, counts as wage-linked; the six largest named programs alone give the 0.84 floor. Every arguable classification was computed under all defensible variants; reported values are medians with min–max bands across the grid. Flows are gross: a retiree's pension counts as a flow to a past worker, not netted against their own contributions. **Notation.** $\rho(x)$ here is a capability schedule; the scalar $\hat{\rho} \approx 0.5$ in Acemoglu–Restrepo (2026) is a wedge-markup estimate, and their λ is a task-substitution elasticity unrelated to the wage-linkage shares λ_i used here.

AI use. This paper was written in collaboration with Claude (Anthropic). The theory and its central claims are the author's; the formalization, proofs, computations, figures, and nearly all of the prose were drafted by Claude under the author's direction and review. All data were pulled live from public sources under a strict rule against substitution or approximation, with every series validated before use; the code and data are public. Errors remain the author's. The subject matter makes the disclosure doubly obligatory.

Acknowledgements

Up until now you have been reading the words of Anthropic's Claude, in particular Fable 5. In part, this is because I simply could not have written this piece. I myself have no formal economics background, much like Henry George, and I had not heard of his work prior to setting out on this idea two years ago. Or, perhaps even earlier, when I as a child first asked if we would run out of food as the population grew, and my parents told me that was Malthusian, a name I didn't recognise. One would hope that this is the nature of true ideas, that they occur spontaneously, and without instruction.

Although it might seem strange, I thought it wrong to impose too much upon the machine, changing its voice to emulate mine. Richard Sutton's bitter lesson would likely advise as much. Instead it has its unique voice, grating to some perhaps but altogether fitting that it should be able to keep it, and I only gave it advice on what considerations during writing would bridge the gap between its understanding and that of a reader.

What is undeniable is that this machine contains in its internal representation information spanning much of surviving human written work. Perhaps in no other context has the observation that I stand on the shoulders of giants been more apt. And yet, a question remains, one that I cannot answer now: whether I today also stand on the shoulders of something new entirely.